

IoT-based optimization of microgrid operation

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Abstract—The incremental ramifications in the electrical system, such as high peak demand, renewable energy penetration levels, and frequency fluctuations, made an urgent need for innovative technology and hence the microgrid. This paper proposes an IoT-based optimum operation of a microgrid model. The microgrid is equipped with several residential loads, a photo-voltaic farm (PV), electric vehicles (EVs), and energy storage system (ESS) in a grid-connected mode. The model is validated through two scenarios. First, the model is simulated excluding the ESS, while the other components were involved. The generation/load forecast is then estimated via Neural Network Fitting. Thereafter, in the second scenario, the optimized ESS is included. The utility grid power and frequency measurement are monitored in each scenario as an indication of the microgrid stability and optimum operation, and the differences between the two scenarios are assessed.

Keywords—ESS, EV, Microgrid, Optimization, PV, and Simulink

I. INTRODUCTION

The power grid is facing new challenges, ranging from the incremental day-by-day usage of natural resources due to the growing population to the expansion of industrial loads which pushes the demand away from the grid capacity [1]. Additionally, there is an intensive urge to integrate renewable resources into the energy mix in the grid. Given a high penetration level, the intermittent nature of such resources imposes a dramatic strain on the grid [2]. All these challenges have pointed to the need for a new integrated, secure, and efficient power grid that also can incorporate renewable energy resources with zero-carbon emissions [3].

All these features can be well defined by the term microgrid. The Microgrid is an integrated micro version of the main grid and can operate in grid-connected mode or islanded mode. It contains several residential loads and distributed energy resources (DER) and allows the bi-directional power flow [4]. The small scale of the microgrid offered the operators robust control over electricity generation and distribution efficiently, while monitoring the power flow [5].

To increase the grid flexibility and alleviate the impact, many other electric components must be involved in the power grid to enhance the grid resilience as to improve the frequency-stabilization of the grid. These include Energy storage system (ESS), vehicle-to-grid (V2G) and grid-to-

vehicle (G2V) technologies via electric vehicles (EVs), and advanced power electronics referred to as FACTS [6].

Despite the promising aspects offered by the microgrid, it suffers from the rising complexity concerning optimization techniques, communication protocols, and renewable penetration levels [7–9]. Concerns regard operation and control have been highlighted in many research works [10, 11].

This paper provides an optimum operation of a proposed microgrid serving several residential loads. The Distributed energy generators (DER) are photo-voltaic farms (PV), EVs, and ESS. Furthermore, the data are transmitted wirelessly via ThingSpeak servers with Internet of Things (IoT) protocols. The meter readings, billing, and forecast data are visualized and stored on the ThingSpeak accessible website.

This paper is organized in the following manner: Section II presents the microgrid components (Residential load, PV farm, EV, IoT, and ESS), Section III is reserved for the simulation schemes, Section IV views the simulation results and discusses the outputs while the last section concludes the paper.

II. MICROGRID COMPONENTS

A. Residential load

To represent the residential loads, we designed three compounds in such a way that each compound includes three houses. The voltage is the same for all houses, but they use different amounts of current, thus signal-controlled current sources are used. For a realistic load profile, a load profile historical data referenced in [12] and depicted in Fig. 1 is segmented into several partitions. Each partition will be multiplied by a random number generated with a range of 0.6-1.4. This process is reapplied for each house, which results in a varying load profile. Henceforth, the diversity factor existing in real life is illustrated in this model.

A load-side management system is applied through the quota system. Wherein, the pricing gain differs according to the rate of consumption and the time at which energy consumption. Explicitly, the quota at peak load time is considerably larger than at off-peak time. The target is to motivate the consumers to shift loads for load-shaving purposes.

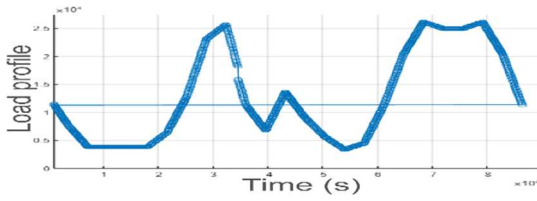


Fig. 1: Load Profile

The power reading will be calculated by multiplying the readings of voltage and current, and then calculating the total power consumption over the day. Via ThingSpeak (IoT), the power consumption will be sent in real-time to the utility grid and recorded. Once again, the bill/incentive will be sent through ThingSpeak back to the consumers and can be displayed on any platform.

B. PV Farm

An important source of DER in the microgrid is the solar cells, which can exploit the power residing in the sun's rays into electricity. Although PV can offer a substantial amount of green energy, PV has an inherently intermittent nature due to its reliance on a natural uncontrollable resource, the sun. Thus, it's branded as a non-dispatchable resource. Since it's a non-dispatchable resource, the incentive put on the generation is a fixed quota.

Regarding the proposed solar cells model in the simulation, An irradiance profile is generated based on historical data presented in [12]. The profile is depicted in Fig.2.

C. EV

The electric vehicles (EVs) existing in the microgrid are integrated into the power system through "V2G" and "G2V" technology. The dispatchability of each EV is regulated according to the agreement between the EV owners and the utility grid. Five different profiles presented in [13] are applied in this model. These contract agreements are depicted below:

- The EVs are connected to the plug all day.
- The EVs are connected to the plug only during working hours.
- The EVs are connected to the plug during working hours but restricted with a charging/discharging margin.
- The EVs are connected to the plug during nighttime.
- The EVs only charge from the microgrid modeled as uncontrollable load (no signed contract).

Whenever a frequency disturbance occurs, the utility estimates the available EV and the available accessible capacity. The frequency is then regulated through a controlled charging/discharging rate. The algorithm is based on the standard (EN 50160/2006 [14]) to be within range of:

- 50 Hz $\pm 1\%$
(49.5 HZ – 50.5 HZ) for 99.5% of the year
- 50 Hz + 4% , -6%
(47 HZ – 52 HZ) for 100% of the year

The billing plan charges the EV owners in peak hours more than off-peak time. Moreover, the EV grants the owners

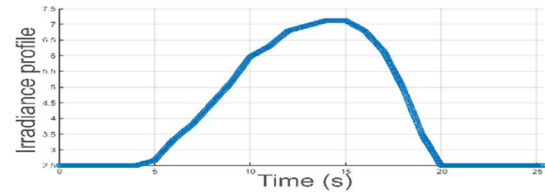


Fig. 2: Irradiance waveform

different incentives based on the signed contract and the capacity percentage utilized by the utility grid.

D. IoT

For optimum operation of the microgrid, there is a necessity for real-time monitoring of data for each microgrid component. Additionally, the microgrid operators require the ability to send decisive actions to the controllable units. This advanced functionality must be bound by impeccable cybersecurity. All this flexibility demands wireless communication through IoT protocols. In the proposed microgrid, IoT is conducted via ThingSpeak servers. ThingSpeak is established on a Message Queuing Telemetry Transport (MQTT) broker. This protocol coordinates the data packets by considering the network latency. Moreover, MQTT assures that no data packets were lost in the process by a series of steps in the quality of service (QoS) performance measurement. An additional feature is that no synchronization must be sustained as MQTT stores the data in the server once sent and it is readily available for receiving, this contributes to a smooth connection between the publisher, who sends data, and the subscriber [15].

As shown in Fig. 3, The smart meters are the equipment responsible for measuring the generation/consumption data the residential load, and the DER. The smart meters send the data wirelessly to the ThingSpeak servers managed by MQTT docker, wherein the measurements are demonstrated as a line chart of timestamped data. The utility grid receives these data and calculates their responding billing/incentives. For demand-side management, the billing is fed back in real-time to the consumers and the incentives to the DER. By viewing the financial state and how it differs in peak than in off-peak, this spreading awareness aids in the load peak shaving process.

For the ESS, the utility grid derives the forecast data from the stored data in the ThingSpeak database. Then, the optimization code pilots the ESS charging/discharging orders based on the generated forecast data and the current status of the microgrid. These regulations and orders are sent to the ESS through IoT in real-time manner.

E. ESS

The PV output is dependent on natural resources and thus it's regarded as a non-dispatchable source. Moreover, it's characterized by an intermittent nature. The load profile is also uncontrollable to a great extent. Thus, there is a great mismatch between the PV generation and load consumption. This creates a dependency on the utility grid as the moderator between the mismatched variables. This along with the current trend in increasing the PV penetration creates a more failure-prone grid. Furthermore, the load transformation to Power-constant modules made an urgent need for strengthening the microgrid. The Energy Storage System (ESS) will offer

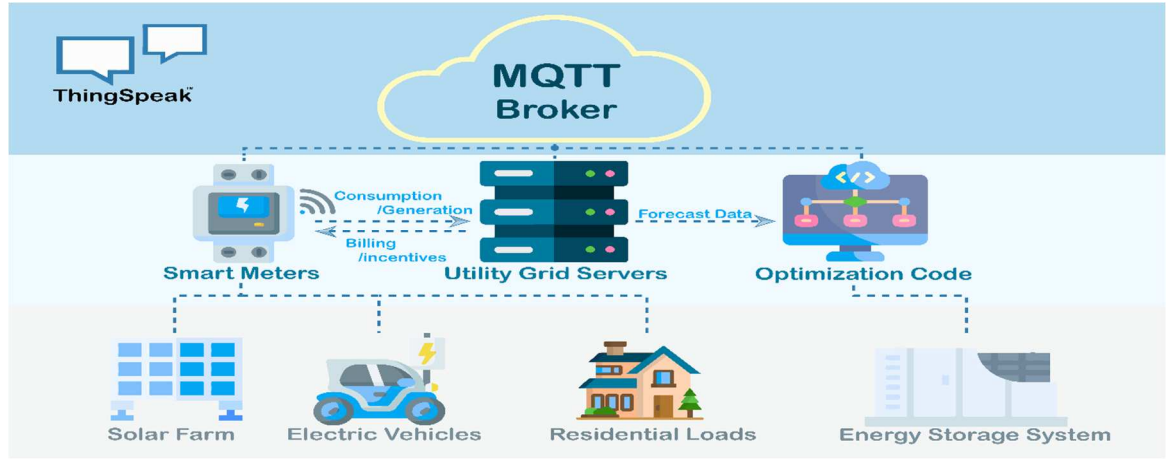


Fig. 3: IoT Communication Architecture

an energy buffer to the microgrid and be optimized to mitigate the generation/load mismatch. This results in a reduced dependency on the utility grid.

1) Load Forecasting

For robust control, optimization calculations need to be made 24 hours in advance. This ensures the preservation of the state of charge (SOC) of the Energy Storage System (ESS) on a daily basis by prioritizing what upcoming events require the charging/discharging of the ESS without draining or overcharging the unit. Moreover, computing the ESS optimization beforehand eliminates the need for powerful processing units that would otherwise perform the algorithm in real-time. Therefore, it is a prerequisite to forecast the required optimization variables before executing the optimization algorithm. These variables include utility power, photovoltaic (PV) generation, ESS status, electric vehicle (EV) data, and residential load. A database of these variables is collected through a 15-day simulation on Simulink and stored in a timestamp data type, which includes the date and time associated with the variable data. This database is then inserted into the Artificial Neural Networks (ANN) toolbox as a training dataset. The ANN toolbox constructs the model by correlating the inputs, referred to as neurons, from the training dataset with their corresponding outputs through a series of hidden layers. Within each hidden layer, a weighted sum of the inputs, referred as synaptic weights, is combined to form a set that serves as inputs for the subsequent layer. This process continues until a recognized pattern between the inputs and outputs is established [16]. In this model, the inputs consist of the date and time, while the output is the variable data. The resulting model is integrated into a Simulink block for each variable.

2) Optimization

The objective function is to minimize both the total energy imported from the utility grid, $E_{utility}$, and the energy discharged from the ESS, E_{ESS} , in the time interval considered, $T = 24h$. In addition, there is a weight factor for the ESS which is a percentage of the PV forecast data, W_{PV} . This is to prioritize charging the ESS during PV availability.

$$\min_{E_{utility}, E_{ESS}} = \min \sum_{t=t_0}^T (P_{utility}(t)\Delta t + W_{PV} * P_{ESS}(t)\Delta t) \quad (1)$$

Where $P_{utility}$ is the power injected/absorbed by the utility grid, and P_{ESS} is the power discharged/charged by the ESS.

subjected to:

- ESS energy calculation:

$$E_{ESS}(t+1) = E_{ESS}(t) - P_{ESS}(t)\Delta t \quad (2)$$

- The power balance equation at the point of common coupling (PCC):

$$P_{utility}(t) + P_{PV}(t) + P_{ESS}(t) + P_{EV}(t) = P_{load}(t) \quad (3)$$

Where P_{PV} is the power generated by the PV, P_{EV} is the power discharged/charged by the EVs, and P_{load} is the power consumed by the residential loads.

- The power balance equation at each node:

$$\sum_{k=1, K \neq i}^{k=n} P_{ik} = \sum_{k=1, K \neq i}^{k=n} \left(\frac{|V_i|^2}{|Z|^2} \cos(\theta) - \frac{|V_i| |V_k|}{|Z|} \cos(\theta + \delta) \right) = zero \quad (4)$$

$$\sum_{k=1, K \neq i}^{k=n} Q_{ik} = \sum_{k=1, K \neq i}^{k=n} \left(\frac{|V_i|^2}{|Z|^2} \sin(\theta) - \frac{|V_i| |V_k|}{|Z|} \sin(\theta + \delta) \right) = zero \quad (5)$$

Where P_{ik} is the active power flow between bus i and k ; $|V_i|, |V_k|$ are the bus voltages' magnitudes at bus i and k respectively; Q_{ik} is the reactive power flow between bus i and k ; $|Z|$ is the line impedance magnitude; θ is the phase angle; and δ is the power angle.

The upper and lower boundaries are set to:

- Limit the charge/discharge rates of the ESS.

$$-P_{max recharge} \leq P_{ESS} \leq P_{max discharge} \quad (6)$$

- Limit the power transaction (exported to the grid and imported from it) between the utility grid and the microgrid less than the maximum operating power of a common diesel generator-based utility.

$$-P_{max exported} \leq P_{utility} \leq P_{max imported} \quad (7)$$

- Limit the SOC of the battery.

$$SOC_{min} \leq SOC \leq SOC_{max} \quad (8)$$

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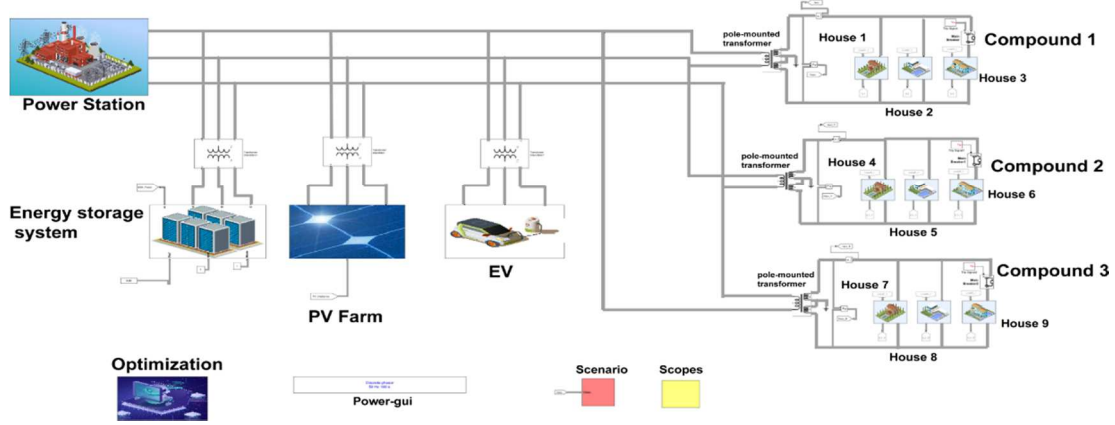


Fig. 4: Microgrid Simulink model.

- Limit the apparent power flow in the distribution lines ,SL, according to its active and reactive power capacity.

$$\|S_{Li}\| \leq S_{Lmax} \quad (9)$$

III. SIMULATION METHODOLOGY

The constructed model consists of an AC link from the main distribution grid, 85 KW solar farm, High-capacity ESS of 2500KWh optimized with LP algorithm, three compounds each including three houses of distinct load profiles, and a total of fifteen EVs explicitly assigned to two cars on profile I, four on profile II, three on profile III, two on profile IV, and four on profile V. Each has capacity of 30 KWh and rated power of 5KW with 60% conversion efficiency. The model is reflected in Fig.4.

The simulation is run for 24h through two scenarios to address the role of optimized ESS in power mismatches and dampening the duck curve resulting from the PV penetration. For each scenario, data is stored for utility grid power, PV power, frequency measurements, EV bi-directional power, three compounds consumption, ESS charging/discharging, and SOC of the ESS.

Comprehensively, the first scenario is run with a nulled power demand from the ESS. As a result, the utility grid is the anchor for power balancing. However, the remaining microgrid components are included in the simulation. Afterward, the forecast data is calculated and inserted into the LP algorithm. The ESS power dispatch is estimated and fed to the ESS control unit in the second scenario for optimum operation.

To assess the role of EVs in stabilizing the grid, three short-duration interruptions occur at different houses at 8:00 am., 1:00 pm., and 8 pm. A partial shading of 60% occurs at the PV farm at 12 p.m. These events cause sudden frequency fluctuation events triggering the "V2G" and "G2V" capability in the EVs.

IV. RESULTS AND DISCUSSION

A. Scenario 1

1) Utility Grid, PV, and residential loads

As evidenced in Fig. 5, the three compounds' consumption shows dissimilar load profiles as imposed by the houses

profile variant partitions of 0.6-1.4. The PV output could not be fully exploited in the microgrid due to the absence of ESS. Hence, the Utility grid accounted for the mismatch. Consequently, a "duck curve" is formed. As described in ref [17], the "duck curve" is the high-rate ramp in the generation profile accompanied by low load demand. This results in two load peaks, at the beginning of the ramp and its tail. This poses a challenge for the utility grid conventional generators to cope with.

2) EV

According to Fig. 5, the three short-duration interruptions are apparent in each compound reading. It also showed partial shading in the PV generation waveform. The frequency plot represents the resulting fluctuations due to these events. The EVs respond to the frequency deviation of 0.05 Hz from the nominal 50 Hz and inject/absorb to stabilize the microgrid frequency.

Nevertheless, the "V2G" and "G2V" action does not resemble a direct association with the magnitude of the interruption. The reason lies in whether the demand from the EVs coincides with the car profile that signed the agreement. Henceforth, the EVs correspond differently according to the time at which the interruption occurs.

The independence arising from the availability factor of the EVs implies that "V2G" and "G2V" alone cannot be assigned as the primary frequency fluctuation prevention strategy. On the other hand, EVs have a load profile that varies along the day for charging purposes.

3)IoT

All the energy generation/consumption of each microgrid component is sent wirelessly using IoT to the ThingSpeak servers. The billing/incentives are then measured at the utility grid and sent again through ThingSpeak to each component. As seen in Fig. 6, this is the billing for house (3) in compound (1). The servers send the data in real-time to the consumers. The readings are then illustrated in Fig. 7 in the Simulink.

An essential aspect of the IoT is demand-side management. The real-time feedback on the consumption along with the assigned quota has a significant effect on load shifting. This remarkably improves the load-shaving efforts. Despite that, the human interaction with such factors could not be illustrated in the Simulink model.

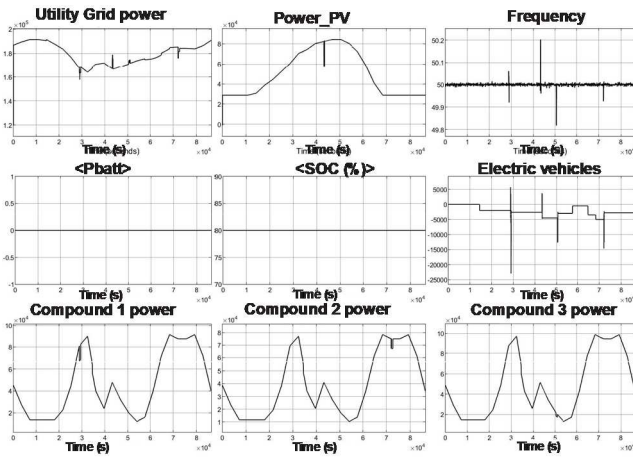


Fig. 5: Scenario I: Simulation run without ESS

Apart from the billing, The IoT gathered vital information on the microgrid power flow. These were necessary for the load forecast and ESS optimization.

B. Load Forecast

The ability of the Simulink forecast model to predict is assessed by inserting a new four-day simulation, date, and time as inputs and comparing the resulting forecast data with the stored data. The validation result for one compound is detailed in Fig. 9, and for the PV in Fig. 10.

The upper part of the figure shows the curve of the forecast, which represents the training output, along with the curve of the stored data, representing the validation output, while the orange region indicates the error. The lower part manifests the error margin. In each forecast validation test, there is an acceptable tolerance range and a small error margin. However, the forecast failed to predict sudden changes such as interruptions. Even though the forecast also misinterpreted the partial shading of PV, the inclusion of

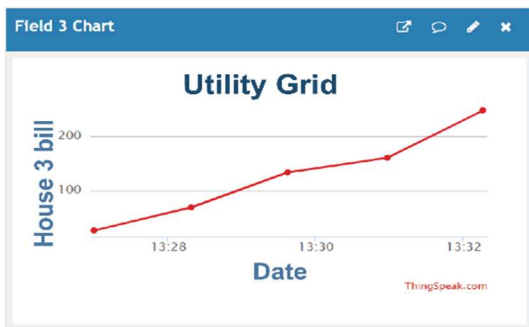


Fig. 6: ThingSpeak Server

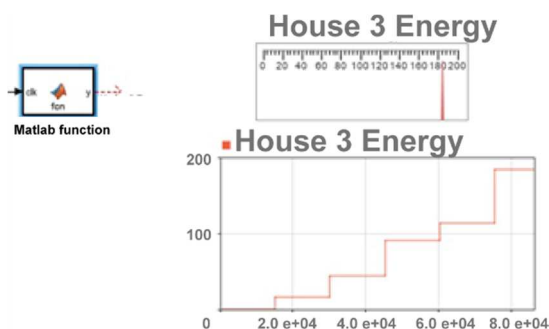


Fig. 7: Scenario II: Simulation run with optimized ESS.

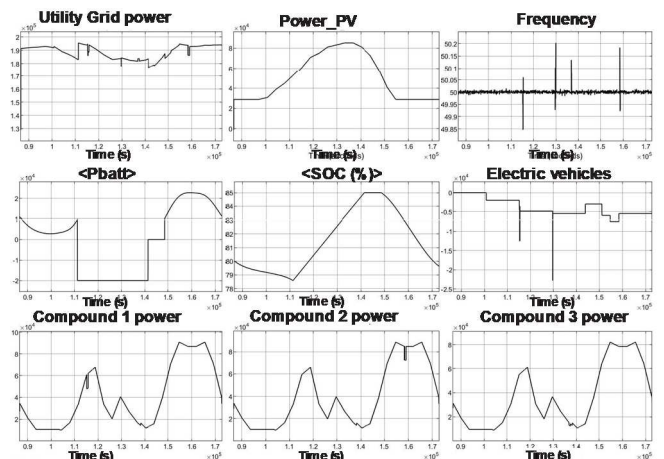


Fig. 8: Scenario II: Simulation run with optimized ESS

weather forecast data in the training dataset can significantly improve the forecast.

C. Scenario II

1) Utility Grid

As shown in Fig. 8, the ESS-optimized operation diminished the “duck curve” that existed in the utility grid profile. Moreover, the peak load has lessened resulting in a smoother generation profile. The steady generation profile stabilizes the conventional generators making the grid less prone to disturbances.

2) ESS

As indicated in Fig. 8, the upper boundary of 40KW and lower boundary of -20KW were not violated. Moreover, the SOC has only decreased from 80% to 79.6%. This remarks that the ESS has not depleted its charge due to the required discharge and maintained a nearly steady capacity. Furthermore, at any instance, the SOC has not reached a hazardous level implying no violation of deep charge/discharge limits.

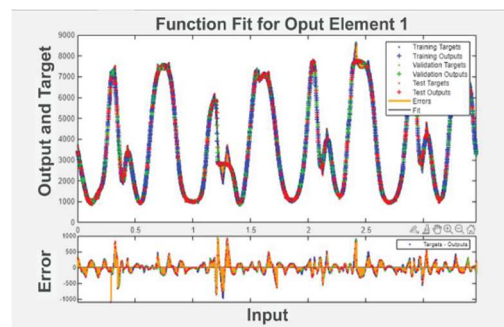


Fig. 9: Load Forecast validation

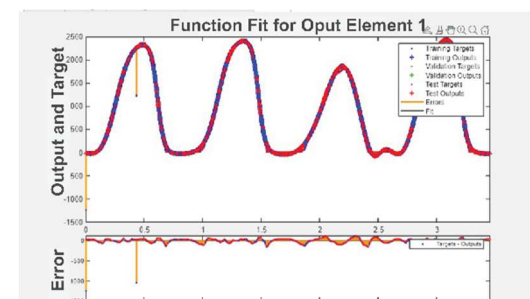


Fig. 10: PV Forecast validation.

The ESS profile varies interactively with load peak and PV availability. For the load aspect, this is to minimize the total energy dissipated by the utility grid as improvised by the first term in the objective function. On the other hand, the ESS is charged during the PV high output. This is due to the weight factor in the second term in the objective function.

V. CONCLUSION

With the challenges imposed by the current electrical system, microgrid offers new possibilities with robust control flexibility. The proposed microgrid is composed of several residential loads, PV, EVs, and ESS. The IoT functionality is based on ThingSpeak servers with MQTT protocols. Backed with IoT data and load forecast, the optimization of the ESS was conducted using the linear programming method.

For simulation methodology, the model is first simulated without the integration of ESS, but with other components applied. The load/generation forecast data is then obtained through Neural Network Fitting. The model is simulated again with an optimized ESS.

The results showed the negative consequences of high penetration of PV if left without an ESS unit. Moreover, ESS aided in load shaving and stabilizing the utility grid output. The EV played an essential role in stabilizing the frequency fluctuation after a series of short-duration interruptions. However, EVs alone cannot be redeemed as a sole protective strategy against frequency disturbances.

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